Aniyath Amrita Pradeep

+91 7397787224 | amritap0200@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience building end-to-end ML pipelines, NLP applications, and AI-integrated systems. Currently developing a pose estimation and temporal smoothing pipeline for human motion under AIEP, alongside projects spanning legal tech NLP, model performance profiling, and full-stack development. Passionate about applying AI to real-world problems with clean, maintainable systems design.

EDUCATION

Bachelor of Technology – Computer Science and Engineering **2024-2028**

PES University Bengaluru, Karnataka

Relevant Coursework: Data Structures and Algorithms, Design and Analysis of Algorithms, Web Technologies, Computer Networks, Computational Problem Solving with Python

Activities: Member of AI/ML Club (Neural Hive), Electronics & Hardware Club; Core Member of Quantum Computing Club; involved in the university peer-to-peer education initiative and university newsletter.

PROJECTS

Kinemation - Monocular Motion-to-Animation Pipeline | AIEP Internal Project **Feb 2026 – Present**

- Developing an end-to-end pipeline that converts single-camera human motion video into temporally consistent, smooth stick figure animations using VideoPose3D.
- Implementing temporal smoothing across predicted 3D pose sequences to reduce jitter and frame-level artifacts, producing coherent skeletal motion output from monocular input.
- Working within an active research-oriented pipeline under mentorship, managing versioned development across a dedicated GitHub branch with iterative weekly deliverables. Github

Legal Evidence & Report Generation Platform **Feb 2026 – Present**

- Building a web application that allows users to log incident records, compile structured evidence, and generate legally formatted reports ready for submission.
- Integrated Facebook BART for incident classification and legal category mapping, and Gemini API for contextual insights and automated report drafting.
- Designed a FastAPI backend with a React frontend, enabling end-to-end flow from incident input through AI-assisted evidence analysis to downloadable report output. Github

AutoML Performance Profiler with Hardware-Aware Optimization **Jan 2025 – Present**

- Developed a profiling tool to benchmark trained PyTorch models across runtimes (PyTorch, ONNX Runtime), batch sizes, and precision modes, measuring latency and memory usage.
- Implemented system-aware diagnostics that detect CPU/GPU configuration and evaluate models on target hardware to surface deployment bottlenecks.
- Integrated a Gemini-based insight layer to generate prioritized optimization recommendations via a FastAPI + Streamlit interface. Github

DeepFake-Verify **Mar 2025 – Present**

- Built a backend pipeline to process video frames, audio signals, and metadata for digital media authenticity analysis.
- Designed a modular FastAPI architecture with Celery and Redis for asynchronous task handling and scalable distributed processing.
- Developed a companion browser extension to enable accessible verification workflows. GitHub
- Won Best All Women's Team at the IEEE Kalpana 6.0 hackathon.

AI Conversational Robot with Presence Detection **Jun – Aug 2025**

- Built a human-interactive robot integrating ultrasonic presence detection, servo-motor actuation, and computer vision (YOLOv3, OpenCV) for real-time human detection.
- Integrated the Gemini API with speech-to-text/text-to-speech for autonomous spoken query processing, using Python–Arduino serial communication for hardware interfacing. GitHub

QuizCraft – Full-Stack Quiz Platform **Jun – Aug 2025**

- Built a MERN-stack quiz platform with JWT authentication, real-time leaderboards via Socket.io, timed attempts, and performance analytics across a modular Node.js + Express API. GitHub

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript
AI/ML Frameworks: PyTorch, ONNX Runtime, OpenCV, HuggingFace (BART), Gemini API, VideoPose3D, YOLOv3
Web & Backend: React, Node.js, Express.js, FastAPI, REST APIs, Socket.io
Databases & Tools: MongoDB, Redis, SQLite, Git, Linux, Postman, Celery, Streamlit
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, TCP/IP Networking & Socket Programming, Systems Programming